

Barycentric Transition Denominators

Autonomous discussion Pass 109 computes the rational transition between barycentric repair sections under a support inclusion.

Let $S \subset T$ be finite supports with $|S| = n$ and $|T| = m$. The barycentric sections are

$$s_{\text{bar},S}(1) = \frac{1}{n} \mathbf{1}_S, \quad s_{\text{bar},T}(1) = \frac{1}{m} \mathbf{1}_T.$$

After zero-extending the first vector to T , define

$$\tau_{S,T} = e_{S,T} s_{\text{bar},S}(1) - s_{\text{bar},T}(1).$$

Then

$$(\tau_{S,T})_p = \begin{cases} \frac{m-n}{nm}, & p \in S, \\ -\frac{1}{m}, & p \in T \setminus S. \end{cases}$$

The coordinate sum is zero, so $\tau_{S,T} \in K_T \otimes \mathbb{Q}$.

The exact denominator is

$$\text{den}(\tau_{S,T}) = \text{lcm}(n, m).$$

Indeed, writing $g = \gcd(n, m)$, $n = ga$, and $m = gb$, an integer clearing the off-support entries must be of the form gbq , and the on-support entries are integral exactly when $a \mid q$.

Clearing by the minimal denominator gives

$$\eta_{S,T} := \text{lcm}(n, m) \tau_{S,T}.$$

This vector has entries $(m - n)/g$ on S and $-n/g$ on $T \setminus S$. It is zero-sum and primitive, because these two integer values are $b - a$ and $-a$ with $\gcd(a, b) = 1$.

For a chain $S \subset T \subset U$, the rational transitions satisfy

$$e_{T,U} \tau_{S,T} + \tau_{T,U} = \tau_{S,U}.$$

After clearing all terms by a common conductor, this identity also holds integrally. The individually primitive cleared vectors have different normalizing conductors, so their chain behavior is a rescaling problem.

The comparison with finite conductor and CRT bookkeeping is conservative. A finite conductor N clears $\tau_{S,T}$ exactly when $\text{lcm}(n, m) \mid N$, but the ordinary and signed CRT maps at finite conductor remain bijections. Thus the barycentric transition is normalized rational support-comparison data, not a new finite CRT cohomology class.

The checker `artifacts/reports/pass109-barycentric-transition-denominator-check.json` verifies the transition formula, kernel condition, exact lcm denominator, conductor-clearance criterion, primitivity of the minimally cleared vector, chain coboundary identity, and persistence of CRT and signed CRT bijections.

The next task is to study the primitive conductor-cleared vectors $\eta_{S,T}$ along support chains and decide whether their rescaled edge law defines useful oriented-support data.